

1 (Zangwill 3.2)

Solution. (a) In charge-free space, Gauss' law implies $\partial_k E_k = 0$. Then since

$$\begin{aligned}\partial_k E_k &= \partial_k (C_k + D_{jk} r_j) \\ &= \cancel{\partial_k C_k} + (\cancel{\partial_k D_{jk}}) r_k + D_{jk} \partial_k r_j \\ &= D_{jk} \delta_{kj} \\ &= D_{kk},\end{aligned}$$

we conclude that $D_{kk} = 0$, i.e. D_{jk} is traceless. If we assume further that there are no time-varying magnetic fields, then Faraday's law implies $\epsilon_{ijk} \partial_j E_k = 0$. Thus

$$\begin{aligned}0 &= \epsilon_{ijk} \partial_j E_k \\ &= \epsilon_{ijk} \partial_j (C_k + D_{\ell k} r_\ell) \\ &= \epsilon_{ijk} \left[\cancel{\partial_j C_k} + (\cancel{\partial_j D_{\ell k}}) r_\ell + D_{\ell k} \partial_j r_\ell \right] \\ &= \epsilon_{ijk} D_{\ell k} \delta_{j\ell} \\ &= \epsilon_{ijk} D_{jk},\end{aligned}$$

and we can act ϵ_{ist} on both sides, then use the identity $\epsilon_{ijk} \epsilon_{ist} = \delta_{js} \delta_{kt} - \delta_{jt} \delta_{ks}$ to obtain

$$\begin{aligned}0 &= (\delta_{js} \delta_{kt} - \delta_{jt} \delta_{ks}) D_{jk} \\ &= D_{st} - D_{ts}.\end{aligned}$$

This shows that D_{jk} is symmetric.

(b) From part (a) and the fact that $\mathbf{E} = -\nabla\varphi$, we have

$$-\begin{pmatrix} \partial_x \varphi \\ \partial_y \varphi \\ \partial_z \varphi \end{pmatrix} = \begin{pmatrix} C_x \\ C_y \\ C_z \end{pmatrix} + \begin{pmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{xy} & D_{yy} & D_{yz} \\ D_{xz} & D_{yz} & D_{zz} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

By inspection, the most general potential leading to this electric field is a degree two polynomial in x , y , and z :

$$\begin{aligned}\varphi(x, y, z) = & - \left[C_x x + C_y y + C_z z + \frac{1}{2} D_{xx} x^2 + \frac{1}{2} D_{yy} y^2 + \frac{1}{2} D_{zz} z^2 \right. \\ & \left. + D_{xy} xy + D_{yz} yz + D_{xz} xz \right] + \text{const.},\end{aligned}$$

subject to the constraint $D_{xx} + D_{yy} + D_{zz} = 0$. □

2 (Jackson 1.5)

Solution. We have

$$\begin{aligned}
 \nabla^2 \Phi &= \nabla^2 \left[\frac{q}{4\pi\epsilon_0} \frac{e^{-\alpha r}}{r} \left(1 + \frac{\alpha r}{2} \right) \right] \\
 &= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{r} \nabla^2 \left[e^{-\alpha r} \left(1 + \frac{\alpha r}{2} \right) \right] + 2 \nabla \left(\frac{1}{r} \right) \cdot \nabla \left[e^{-\alpha r} \left(1 + \frac{\alpha r}{2} \right) \right] + e^{-\alpha r} \left(1 + \frac{\alpha r}{2} \right) \nabla^2 \left(\frac{1}{r} \right) \right\} \\
 &= \frac{q}{4\pi\epsilon_0} \left[\frac{\alpha^3 e^{-\alpha r}}{2} - 4\pi \delta^3(r) e^{-\alpha r} \left(1 + \frac{\alpha r}{2} \right) \right] \\
 &= -\frac{q \delta^3(r)}{\epsilon_0} + \frac{q \alpha^3 e^{-\alpha r}}{8\pi\epsilon_0},
 \end{aligned}$$

where in the second line, we use the product rule for the Laplacian,

$$\nabla^2(fg) = f(\nabla^2 g) + 2(\nabla f) \cdot (\nabla g) + g(\nabla^2 f).$$

In the third line, we use Mathematica to compute and combine some of the derivatives, and we use the identity $\nabla^2(1/r) = -4\pi\delta^3(r)$. Lastly, in the fourth line, we replace the expression multiplied to $\delta^3(r)$ by its value at $r = 0$. Since $\nabla^2 \Phi = -\rho/\epsilon_0$, we conclude that

$$\boxed{\rho(r) = q \delta^3(r) - \frac{q \alpha^3 e^{-\alpha r}}{8\pi}}.$$

Physically, this corresponds to a positive point charge q at $r = 0$ embedded in a spherically symmetric cloud of negative charge. This cloud is densest at the origin, and dies off exponentially (with characteristic length scale α^{-1}) as r increases.

Furthermore, since

$$\begin{aligned}
 Q_{\text{cloud}} &= \int_0^\infty 4\pi r^2 \left(-\frac{q \alpha^3 e^{-\alpha r}}{8\pi} \right) dr = -q \int_0^\infty \frac{\alpha^3 r^2 e^{-\alpha r}}{2} dr \\
 &= -q,
 \end{aligned}$$

it follows that the total charge of the system is zero, as we expect. In fact, the charge density of the cloud dies off so quickly that most of the total charge is contained within a few Bohr radii of the origin. This means that at a macroscopic scale, the hydrogen atom is effectively neutral; it is only when $r \sim a_0$ that there is an appreciable electric field. $\square$

3 (Jackson 1.6)

Solution. (a) To a good approximation, the charge on each sheet will be spread evenly (with charge density per surface area $\pm\sigma$), and since the sheets are assumed large, the electric field near each sheet will point along the normal, by symmetry. Then the flux through a squat Gaussian surface of area a enclosing one of the sheets is

$$\Phi = 2Ea.$$

By Gauss' law, we also have

$$\Phi = \frac{\sigma a}{\epsilon_0}.$$

Between these two expressions, we find

$$E = \frac{\sigma}{2\epsilon_0}.$$

The field in between the sheets is twice this (since one sheet has field lines pointing inward, and the other has lines pointing outward), and then the potential difference is given by

$$V = \frac{\sigma d}{\epsilon_0}.$$

Finally, the capacitance of the setup is

$$C = \frac{\sigma A}{\sigma d/\epsilon_0} = \frac{\epsilon_0 A}{d}.$$

(b) Say that the inner shell has charge density per surface area σ . By symmetry, the flux through a spherical Gaussian surface of radius r (where $a \leq r \leq b$) is given by

$$\Phi = E(4\pi r^2),$$

and by Gauss' law,

$$\Phi = \frac{4\pi a^2 \sigma}{\epsilon_0}.$$

Thus, in the region between the shells, the electric field is

$$E = \frac{\sigma a^2}{\epsilon_0 r^2}.$$

The potential difference between the shells is then

$$V = - \int_b^a \frac{\sigma a^2}{\epsilon_0 r^2} dr = \frac{\sigma a^2}{\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{\sigma a^2}{\epsilon_0} \frac{b-a}{ab},$$

which means that the capacitance of the setup is

$$C = \frac{4\pi a^2 \sigma}{\sigma a^2/\epsilon_0} \frac{ab}{b-a} = \frac{4\pi \epsilon_0 ab}{b-a}.$$

(c) Say that the inner cylinder has charge density per unit area σ . By symmetry, the flux through a cylindrical Gaussian surface of radius r , where $a \leq r \leq b$, is

$$\Phi = E(2\pi rL),$$

and by Gauss' law

$$\Phi = \frac{2\pi aL\sigma}{\epsilon_0},$$

which gives

$$E = \frac{\sigma a}{\epsilon_0 r}.$$

Then the potential difference between the cylinders is

$$V = - \int_b^a \frac{\sigma a}{\epsilon_0 r} dr = \frac{\sigma a}{\epsilon_0} \ln\left(\frac{b}{a}\right).$$

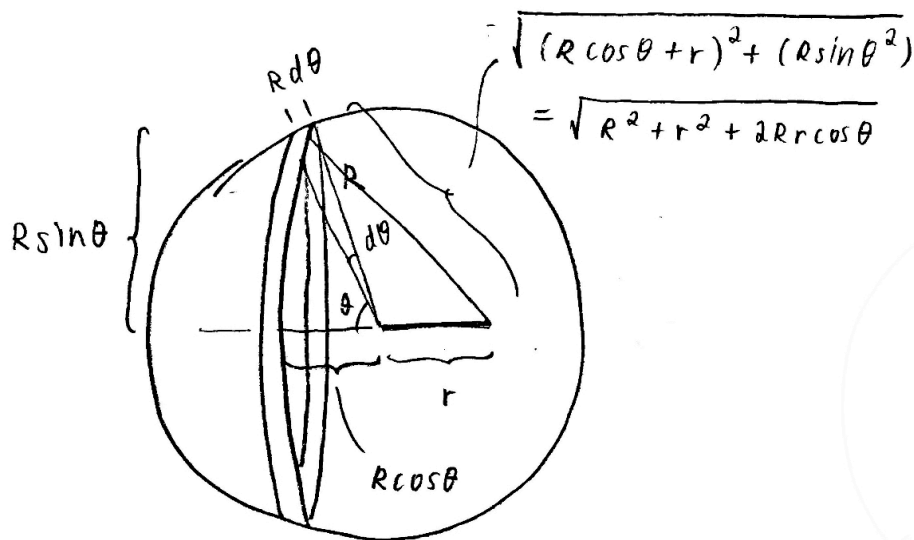
Finally, the capacitance of this setup is

$$C = \frac{2\pi aL\sigma}{(\sigma a/\epsilon_0) \ln(b/a)} = \frac{2\pi\epsilon_0 L}{\ln(b/a)}.$$

In all of these systems, you can increase C by either enlarging the physical dimensions of the system or by decreasing the separation of the conductors. $\square$

4 (Zangwill 3.14)

Solution. (a)



We regard the shell as being composed of many small patches, each of which we can treat as a point carrying a charge of $dq = \sigma dA$. The potential due to a thin ring of these patches, as indicated in the drawing, is

$$d\varphi = \frac{\sigma 2\pi R^2 \sin \theta d\theta}{4\pi\epsilon_0} (R^2 + r^2 + 2Rr \cos \theta)^{-(1+\epsilon)/2}.$$

Thus, we have

$$\begin{aligned} \varphi(r) &= \frac{\sigma R^2}{2\epsilon_0} \int_0^\pi \frac{\sin \theta d\theta}{(R^2 + r^2 + 2Rr \cos \theta)^{(1+\epsilon)/2}} \\ &= \frac{\sigma R}{4\epsilon_0 r} \int_{(R-r)^2}^{(R+r)^2} u^{-(1+\epsilon)/2} du, \end{aligned}$$

where we make the substitution $u = R^2 + r^2 + 2Rr \cos \theta$. Evaluating the integral, we get

$$\boxed{\varphi(r) = \frac{\sigma R}{2\epsilon_0 r} \frac{1}{1-\epsilon} \left(|R+r|^{1-\epsilon} - |R-r|^{1-\epsilon} \right),}$$

so that in the Coulomb limit $\epsilon = 0$, the potential takes a constant value of $\sigma R/\epsilon_0$ inside the shell. In this limit, the electric field is zero inside the shell, as we expect.

(b) From part (a), we have

$$\frac{\varphi(R) - \varphi(r)}{\varphi(R)} = \frac{\frac{(2R)^{1-\epsilon}}{R} - \frac{[(R+r)^{1-\epsilon} - (R-r)^{1-\epsilon}]}{r}}{\frac{(2R)^{1-\epsilon}}{R}}.$$

Expecting a small deviation, if ever, from the Coulomb potential, we expand around $\epsilon = 0$. Using the fact that

$$a^x = e^{x \ln a} = 1 + x \ln a + \frac{1}{2}(x \ln a)^2 + \cdots,$$

we get, discarding terms of order ϵ^2 and higher,

$$(R + r)^{1-\epsilon} = (R + r)[1 - \epsilon \ln(R + r)],$$

$$(R - r)^{1-\epsilon} = (R - r)[1 - \epsilon \ln(R - r)].$$

Thus

$$(2R)^{1-\epsilon} = 2R(1 - \epsilon \ln 2R),$$

$$(R + r)^{1-\epsilon} - (R - r)^{1-\epsilon} = 2r - \epsilon R \ln \frac{R + r}{R - r} - \epsilon r \ln(R^2 - r^2),$$

and we obtain, still working up to order ϵ ,

$$\begin{aligned} \frac{\varphi(R) - \varphi(r)}{\varphi(R)} &= \frac{2(1 - \epsilon \ln 2R) - 2 + \frac{\epsilon R}{r} \ln \frac{R + r}{R - r} + \epsilon \ln(R^2 - r^2)}{2(1 - \epsilon \ln 2R)} \\ &= \left[1 - \epsilon \ln 2R - 1 + \frac{\epsilon R}{2r} \ln \frac{R + r}{R - r} + \frac{\epsilon}{2} \ln(R^2 - r^2) \right] (1 + \epsilon \ln 2R) \\ &= \frac{\epsilon}{2} \left[\frac{R}{r} \ln \frac{R + r}{R - r} - 2 \ln 2R + \ln(R^2 - r^2) \right] \\ &= \frac{\epsilon}{2} \left[\frac{R}{r} \ln \frac{R + r}{R - r} - \ln \frac{4R^2}{R^2 - r^2} \right]. \end{aligned}$$

This is what we wanted to show. □

5 (Zangwill 3.15)

Solution. (a) The total charge contained in the system is

$$Q = \frac{4}{3}\pi(R^3 - a^3)\rho,$$

where ρ is some constant. Since there is no electric field inside the inner sphere (it encloses zero charge), we have

$$U_E = \frac{1}{2} \epsilon_0 \int d^3r |\mathbf{E}|^2 = \frac{1}{2} \epsilon_0 \left(\int_a^R dr 4\pi r^2 |\mathbf{E}_{a \leq r \leq R}|^2 + \int_R^\infty dr 4\pi r^2 |\mathbf{E}_{r \geq R}|^2 \right).$$

By Gauss' law,

$$\begin{aligned} |\mathbf{E}_{a \leq r \leq R}| &= \frac{\frac{4}{3}\pi(r^3 - a^3)\rho}{4\pi\epsilon_0 r^2} \\ &= \frac{Q}{4\pi\epsilon_0(R^3 - a^3)} \left(r - \frac{a^3}{r^2} \right) \\ &\equiv \frac{Q}{4\pi\epsilon_0 R^3(1 - x^3)} \left(r - \frac{a^3}{r^2} \right), \quad (x \equiv a/R) \\ |\mathbf{E}_{r \geq R}| &= \frac{Q}{4\pi\epsilon_0 r^2}. \end{aligned}$$

Thus, the contribution to the potential energy from the outer region is

$$\begin{aligned} \frac{1}{2} \epsilon_0 \int_R^\infty dr 4\pi r^2 \frac{Q^2}{16\pi^2 \epsilon_0^2 r^4} &= \frac{Q^2}{8\pi\epsilon_0} \int_R^\infty \frac{dr}{r^2} \\ &= \frac{Q^2}{8\pi\epsilon_0 R}, \end{aligned}$$

and the contribution from the inner region is

$$\begin{aligned} \frac{1}{2} \epsilon_0 \int_a^R dr 4\pi r^2 \frac{Q^2}{16\pi^2 \epsilon_0^2 R^6(1 - x^3)^2} \left(r^2 - \frac{2a^3}{r} + \frac{a^6}{r^4} \right) \\ &= \frac{Q^2}{8\pi\epsilon_0 R^6(1 - x^3)^2} \int_a^R dr \left(r^4 - 2a^3 r + \frac{a^6}{r^2} \right) \\ &= \frac{Q^2}{8\pi\epsilon_0 R^6(1 - x^3)^2} \left[\frac{1}{5} R^5 - \frac{1}{5} a^5 - a^3 R^2 + a^5 - \frac{a^6}{R} + a^5 \right] \\ &= \frac{Q^2}{8\pi\epsilon_0 R(1 - x^3)^2} \frac{-5x^6 + 9x^5 - 5x^3 + 1}{5}, \end{aligned}$$

so that the potential energy of the entire system is

$$U_E = \frac{Q^2}{8\pi\epsilon_0 R} \left(1 + \frac{-5x^6 + 9x^5 - 5x^3 + 1}{5(1 - x^3)^2} \right).$$

Now we check the limiting forms of this expression. When $x = 0$, which corresponds to a uniform sphere of charge, this gives

$$U_E = \frac{3Q^2}{20\pi\epsilon_0 R},$$

which is correct (we derived this in the lectures). Meanwhile,

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{-5x^6 + 9x^5 - 5x^3 + 1}{5(1 - x^3)^2} &= \lim_{x \rightarrow 1} \frac{-30x^5 + 45x^4 - 15x^2}{10(1 - x^3)3x^2} \\ &= \lim_{x \rightarrow 1} \frac{-150x^4 + 180x^3 - 30x}{-90x^4 + 60x(1 - x^3)} \\ &= 0, \end{aligned}$$

where we've applied L'Hopital's rule twice. Therefore, when $x = 1$, which corresponds to a thin shell of charge, our expression gives

$$U_E = \frac{Q^2}{8\pi\epsilon_0 R},$$

which is also correct.

(b) To minimize the potential energy with respect to x , we first look for x satisfying

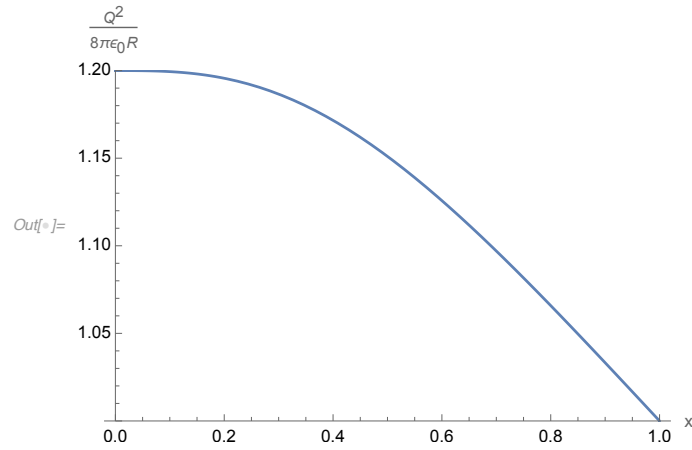
$$\frac{dU_E}{dx} = 0.$$

Using Mathematica to compute the derivative, we find that this holds if and only if

$$\frac{x^2(1 + 3x + x^2)}{1 + x + x^2} = 0.$$

Since $x \equiv a/R$ is real and nonnegative, the solution is $x = 0$. Then since U_E is continuous on the interval $[0, 1]$, it attains its minimum value either where $dU_E/dx = 0$ or at one of the endpoints of the interval.


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In[ ]:= Plot[1 +  $\frac{-5 x^6 + 9 x^5 - 5 x^3 + 1}{5 (1 - x^3)^2}$ , {x, 0, 1}, PlotRange -> {1, 6/5}, AxesLabel -> {"x", " $\frac{Q^2}{8\pi\epsilon_0 R}$ "}]
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Since we've already evaluated U_E at all of these points, we conclude that

$$U_E \text{ attains a minimum value of } Q^2/8\pi\epsilon_0 R \text{ at } x = 1.$$

Interestingly, for the system described in the problem statement, the way to minimize U_E for fixed Q and R is to concentrate all of the charge in a thin shell located at $r = R$. $\square$

6 (Zangwill 3.22)

Solution. (a) Integrating out over rings of radius ρ and thickness $d\rho$, we have

$$\begin{aligned} Q &= \int_0^\infty d\rho 2\pi\rho \sigma \\ &= \int_0^\infty d\rho 2\pi\rho \left(-\frac{qd}{2\pi(\rho^2 + s^2)^{3/2}} \right) \\ &= -qd \int_0^\infty d\rho \frac{\rho}{(\rho^2 + s^2)^{3/2}} \\ &= -\frac{qd}{2} \int_{s^2}^\infty \frac{du}{u^{3/2}} \\ &= \frac{qd}{\sqrt{u}} \Big|_{s^2}^\infty \end{aligned}$$

where in the fourth line we make the substitution $u = \rho^2 + s^2$. Evaluating the expression in the last line, we find that the total charge is

$$Q = -\frac{qd}{s}.$$

(b) Without loss of generality, we can assume that $z > 0$, since the setup is invariant with respect to the reflection $z \rightarrow -z$. Now, the potential $d\varphi$ due to one of the thin rings at a point P on the z -axis is (see the drawing)

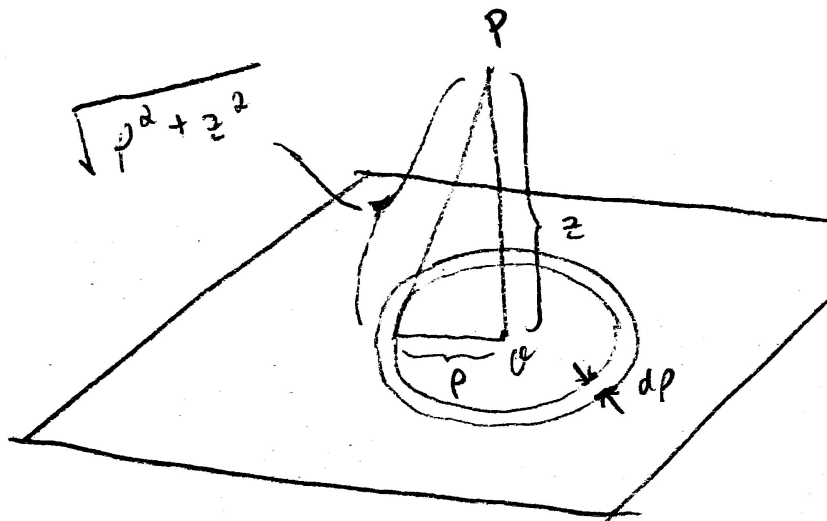
$$d\varphi = -\frac{qd}{4\pi\epsilon_0} \frac{\rho d\rho}{(\rho^2 + s^2)^{3/2}(\rho^2 + z^2)^{1/2}}.$$

Integrating out to infinity and making the substitution $u = (\rho^2 + z^2)^{1/2}$, we get

$$\varphi = -\frac{qd}{4\pi\epsilon_0} \int_z^\infty \frac{du}{(u^2 + s^2 - z^2)^{3/2}}.$$

Annoyingly, this seems to require case-by-case treatment for the three possibilities $s > z$, $s = z$, and $s < z$. We'll consider each one in turn, but it would be nice if there were some argument to avoid this. First, if $s > z$, then we can define $a^2 \equiv s^2 - z^2$ and make the substitution $u = a \tan \theta$ to get

$$\begin{aligned} \varphi &= -\frac{qd}{4\pi\epsilon_0} \int_{\tan^{-1}(z/a)}^{\pi/2} \frac{a \sec^2 \theta d\theta}{a^3 \sec^3 \theta} \\ &= -\frac{qd}{4\pi a^2 \epsilon_0} \sin \theta \Big|_{\tan^{-1}(z/a)}^{\pi/2} \\ &\equiv -\frac{qd}{4\pi\epsilon_0} \frac{1}{s^2 - z^2} \left(1 - \frac{z}{s} \right) \\ &= \frac{Q}{4\pi\epsilon_0(z + s)}. \end{aligned}$$



Meanwhile, if $s < z$, we can define $a^2 \equiv z^2 - s^2$ and then let $u = a \sec \theta$ to obtain

$$\begin{aligned}
 \varphi &= -\frac{qd}{4\pi\epsilon_0} \int_{\sec^{-1}(z/a)}^{\pi/2} \frac{a \sec \theta \tan \theta d\theta}{a^3 \tan^3 \theta} \\
 &= -\frac{qd}{4\pi\epsilon_0} \int_{\sec^{-1}(z/a)}^{\pi/2} \frac{\cot \theta \csc \theta d\theta}{a^2} \\
 &= \frac{qd}{4\pi\epsilon_0} \frac{\csc \theta}{a^2} \Big|_{\sec^{-1}(z/a)}^{\pi/2} \\
 &= \frac{qd}{4\pi\epsilon_0 a^2} \left(1 - \frac{z}{\sqrt{z^2 - a^2}} \right) \\
 &= \frac{qd}{4\pi\epsilon_0} \frac{s - z}{s(z^2 - s^2)} \\
 &= \frac{Q}{4\pi\epsilon_0(z + s)},
 \end{aligned}$$

where in the second-to-last line we used the definition of a^2 . Finally, when $s = z$ the integral is easily shown to be

$$\varphi = -\frac{qd}{8\pi\epsilon_0 z^2} \equiv \frac{Q}{4\pi\epsilon_0(z + s)}.$$

Thus, the potential $\varphi(z)$ produced along the z -axis by this charge distribution is given by

$$\varphi(z) = \frac{Q}{4\pi\epsilon_0(z + s)}.$$

This is precisely the potential due to a point charge Q located a distance s away from the opposite side of the plane. □

1 (Zangwill 4.3)

Solution. (a) Consider the electric dipole shown in Fig. 4.5 of Zangwill, which has zero total charge $Q = 0$ and dipole moment $\mathbf{p} = q\mathbf{b}$. The point electric dipole is the $s \rightarrow 0$ limit of this arrangement, with corresponding charge density

$$\begin{aligned}\rho_D(\mathbf{r}) &= \lim_{s \rightarrow 0} \left[\frac{q}{s} \delta(\mathbf{r} - \mathbf{r}_0 - s\mathbf{b}) - \frac{q}{s} \delta(\mathbf{r} - \mathbf{r}_0) \right] \\ &= q \lim_{s \rightarrow 0} \underbrace{\left[\frac{\delta(\mathbf{r} - \mathbf{r}_0 - s\mathbf{b}) - \delta(\mathbf{r} - \mathbf{r}_0)}{s} \right]}_{\equiv -\mathbf{b} \cdot \nabla \delta(\mathbf{r} - \mathbf{r}_0)} \\ &= -\mathbf{p} \cdot \nabla \delta(\mathbf{r} - \mathbf{r}_0),\end{aligned}$$

where in the second line we use the definition of the directional derivative, and in the third line we use the fact that $\mathbf{p} = q\mathbf{b}$.

(b) We have

$$\begin{aligned}\varphi(\mathbf{r}) &\stackrel{1}{=} \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{\rho_D(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \\ &\stackrel{2}{=} -\frac{\mathbf{p}}{4\pi\epsilon_0} \cdot \int d^3r' \frac{\nabla' \delta(\mathbf{r}' - \mathbf{r}_0)}{|\mathbf{r} - \mathbf{r}'|} \\ &\stackrel{3}{=} -\frac{\mathbf{p}}{4\pi\epsilon_0} \cdot \int d^3r' \left[\nabla' \left(\frac{\delta(\mathbf{r}' - \mathbf{r}_0)}{|\mathbf{r} - \mathbf{r}'|} \right) - \delta(\mathbf{r}' - \mathbf{r}_0) \nabla' \frac{1}{|\mathbf{r} - \mathbf{r}'|} \right] \\ &\stackrel{4}{=} -\frac{\mathbf{p}}{4\pi\epsilon_0} \cdot \left\{ \int dS' \hat{\mathbf{n}} \frac{\delta(\mathbf{r}' - \mathbf{r}_0)}{|\mathbf{r} - \mathbf{r}'|} - \int d^3r' \delta(\mathbf{r}' - \mathbf{r}_0) \nabla' \frac{1}{|\mathbf{r} - \mathbf{r}'|} \right\} \\ &\stackrel{5}{=} \frac{\mathbf{p}}{4\pi\epsilon_0} \cdot \int d^3r' \delta(\mathbf{r}' - \mathbf{r}_0) \nabla' \frac{1}{|\mathbf{r} - \mathbf{r}'|} \\ &\stackrel{6}{=} -\frac{\mathbf{p}}{4\pi\epsilon_0} \cdot \int d^3r' \delta(\mathbf{r}' - \mathbf{r}_0) \nabla \frac{1}{|\mathbf{r} - \mathbf{r}'|} \\ &\stackrel{7}{=} -\frac{1}{4\pi\epsilon_0} \mathbf{p} \cdot \nabla \frac{1}{|\mathbf{r} - \mathbf{r}_0|},\end{aligned}$$

as we wanted to show. The third line uses the product rule for ∇ , the fourth line uses the divergence theorem, and the fifth line uses the fact that $\delta(\mathbf{r}' - \mathbf{r}_0)$ is zero everywhere on the surface. Finally, line six uses the fact that

$$\frac{\partial}{\partial x} f(x - x') = -\frac{\partial}{\partial x'} f(x - x')$$

to remove the prime from ∇ at the cost of an overall minus sign. □

2 (Zangwill 4.16)

Solution. (a) This is a direct calculation using the electrostatic potential in (4.54):

$$\begin{aligned}
 \rho_Q(\mathbf{r}) &= -\epsilon_0 \nabla^2 \varphi(\mathbf{r}) \\
 &= -\frac{1}{4\pi} \nabla^2 \left[(\mathbf{c} \cdot \nabla)(\mathbf{p} \cdot \nabla) \frac{1}{|\mathbf{r} - \mathbf{r}_0|} \right] \\
 &= -\frac{1}{4\pi} \nabla^2 \left[c_k \nabla_k \left(p_l \nabla_l \frac{1}{|\mathbf{r} - \mathbf{r}_0|} \right) \right] \\
 &= -\frac{1}{4\pi} \nabla^2 \left[\frac{1}{2} (c_k p_l + c_l p_k) \nabla_k \nabla_l \frac{1}{|\mathbf{r} - \mathbf{r}_0|} \right] \\
 &= -\frac{1}{4\pi} Q_{kl} \nabla_k \nabla_l \nabla^2 \frac{1}{|\mathbf{r} - \mathbf{r}_0|} \\
 &= Q_{ij} \nabla_i \nabla_j \delta(\mathbf{r} - \mathbf{r}_0).
 \end{aligned}$$

We can move the components c_k and p_l through the derivative operators in lines four and five because they are constants. The fifth line uses the equality of mixed partial derivatives, and the sixth line uses the fact that $Q_{kl} = \frac{1}{2}(c_k p_l + c_l p_k)$ for this potential. Finally, the last line uses the identity $\nabla^2(1/|\mathbf{r} - \mathbf{r}_0|) = -4\pi\delta(\mathbf{r} - \mathbf{r}_0)$.

(b) Using the result from the previous part, we have

$$\begin{aligned}
 \mathbf{F} &= \int d^3r' \rho_Q(\mathbf{r}') \mathbf{E}(\mathbf{r}') \\
 &= Q_{ij} \int d^3r' \left(\nabla'_i \nabla'_j \delta(\mathbf{r}' - \mathbf{r}_0) \right) \mathbf{E}(\mathbf{r}') \\
 &= Q_{ij} \partial_i \partial_j \mathbf{E}(\mathbf{r}_0).
 \end{aligned}$$

This calculation looks like it is less careful with the use of the delta function, but the answer turns out to be correct.¹

¹You can *define* partial differentiation of a distribution u by

$$(\partial_k u)(\phi) = -u(\partial_k \phi)$$

for smooth functions ϕ with compact support. It follows from iteration that if α is a multi-index, so that $|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_n$, then one has $(\partial_\alpha u)(\phi) = (-1)^{|\alpha|} u(\partial_\alpha \phi)$. In particular, for the Dirac delta distribution centered at $a \in \mathbb{R}^n$, defined by

$$\delta_a(\phi) = \phi(a),$$

one obtains

$$(\partial_i \partial_j \delta_a)(\phi) = \delta_a(\partial_i \partial_j \phi) = \partial_i \partial_j \phi|_{x=a}$$

which is what we used in part (b). Incidentally, we could have used the same reasoning to do Problem 1(b) in about four fewer lines, but I didn't know about these results when I typed up that solution.

(c) We have

$$\begin{aligned}
 \mathbf{N} &= \int d^3r' \mathbf{r}' \times \mathbf{E}(\mathbf{r}') \rho_Q(\mathbf{r}') \\
 &= \int d^3r' \mathbf{r}' \times \mathbf{E}(\mathbf{r}') Q_{ij} \nabla'_i \nabla'_j \delta(\mathbf{r}' - \mathbf{r}_0) \\
 &= Q_{ij} \nabla_i \nabla_j (\mathbf{r} \times \mathbf{E}(\mathbf{r})) \Big|_{\mathbf{r}=\mathbf{r}_0}
 \end{aligned}$$

using the result from the footnote. Now look at the components:

$$\begin{aligned}
 N_l &= Q_{ij} \nabla_i \nabla_j (\epsilon_{lmn} r_m E_n) \\
 &= Q_{ij} \epsilon_{lmn} \nabla_i (\delta_{jm} E_n + r_m \nabla_j E_n) \\
 &= Q_{ij} \epsilon_{lmn} (\delta_{jm} \nabla_i E_n + \delta_{im} \nabla_j E_n + r_m \nabla_i \nabla_j E_n) \\
 &= Q_{im} \epsilon_{lmn} \nabla_i E_n + Q_{mj} \epsilon_{lmn} \nabla_j E_n + \epsilon_{lmn} r_m Q_{ij} \partial_i \partial_j E_n \\
 &= 2\epsilon_{lmn} Q_{mj} \nabla_j E_n + \epsilon_{lmn} r_m Q_{ij} \partial_i \partial_j E_n \\
 &= 2[(\mathbf{Q} \cdot \nabla) \times \mathbf{E}]_l + (\mathbf{r} \times \mathbf{F})_l
 \end{aligned}$$

In the second-to-last line, we relabel a bound index and use the symmetry of the quadrupole tensor to combine the first two terms on the fourth line. In the last line, we use the expression for $\mathbf{F}$ from part (b), as well as the definition $(\mathbf{Q} \cdot \nabla)_i = Q_{ij} \nabla_j$. This shows that

$$\mathbf{N} = 2(\mathbf{Q} \cdot \nabla) \times \mathbf{E} + \mathbf{r} \times \mathbf{F},$$

where the right-hand side is to be evaluated at $\mathbf{r} = \mathbf{r}_0$.

(d) We have

$$\begin{aligned}
 V_E(\mathbf{r}) &= \int d^3r' \varphi(\mathbf{r}') \rho_Q(\mathbf{r}') \\
 &= \int d^3r' \varphi(\mathbf{r}') Q_{ij} \nabla'_i \nabla'_j \delta(\mathbf{r}' - \mathbf{r}_0) \\
 &= Q_{ij} \nabla_i \nabla_j \varphi(\mathbf{r}) \Big|_{\mathbf{r}=\mathbf{r}_0} \\
 &= -Q_{ij} \partial_i \partial_j E_j(\mathbf{r}_0),
 \end{aligned}$$

using the result from the footnote, as well as $\mathbf{E} = -\nabla\varphi$. □

3 (Zangwill 4.20)

Solution. (a) The exterior spherical multipole moments are

$$\begin{aligned}
 A_{lm} &= \frac{4\pi}{2l+1} \int d^3r' \rho(\mathbf{r}') (r')^l Y_{lm}^*(\Omega') \\
 &= \frac{4\pi}{2l+1} \int d^3r' (\sigma_0 \sin \theta' \cos \phi' \delta(r' - R)) (r')^l Y_{lm}^*(\Omega') \\
 &= \frac{4\pi\sigma_0}{2l+1} \int_0^\infty dr' (r')^{l+2} \delta(r' - R) \int d\Omega' \sin \theta' \cos \phi' Y_{lm}^*(\Omega') \\
 &= \frac{4\pi\sigma_0}{2l+1} R^{l+2} \int d\Omega' \sin \theta' \cos \phi' Y_{lm}^*(\Omega').
 \end{aligned}$$

The key step for the angular integral is recognizing that the charge density can be expressed as a sum of spherical harmonics, which allows us to exploit orthonormality. In particular,

$$Y_{11}(\theta', \phi') - Y_{1,-1}(\theta', \phi') = -\sqrt{\frac{3}{2\pi}} \sin \theta' \cos \phi', \quad (1)$$

which implies that

$$\begin{aligned}
 A_{lm} &= -\frac{4\pi\sigma_0}{2l+1} R^{l+2} \sqrt{\frac{2\pi}{3}} \int d\Omega' (Y_{11}(\theta', \phi') - Y_{1,-1}(\theta', \phi')) Y_{lm}^*(\Omega') \\
 &= -\frac{4\pi\sigma_0}{2l+1} R^{l+2} \sqrt{\frac{2\pi}{3}} (\delta_{1l}\delta_{1m} - \delta_{1l}\delta_{-1,m}).
 \end{aligned}$$

Thus, there are only two nonzero multipole moments, namely A_{11} and $A_{1,-1}$. Evaluating these and then simplifying gives

$$\boxed{A_{11} = -\left(\frac{32\pi^3}{27}\right)^{1/2} \sigma_0 R^3, \quad A_{1,-1} = -A_{11}.}$$

(b) Using the exterior multipole expansion and the fact that $\sin \theta \cos \phi = x/r$, we get

$$\begin{aligned}
 \varphi(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} \frac{Y_{lm}(\Omega)}{r^{l+1}} \\
 &= \frac{1}{4\pi\epsilon_0} \left(\frac{A_{11}Y_{11}(\Omega)}{r^2} + \frac{A_{1,-1}Y_{1,-1}(\Omega)}{r^2} \right) \\
 &= \frac{\sigma_0 R^3}{4\pi\epsilon_0} \left(\frac{32\pi^3}{27} \right)^{1/2} \left(\frac{-Y_{11}(\Omega) + Y_{1,-1}(\Omega)}{r^2} \right) \\
 &= \frac{\sigma_0 R^3}{4\pi\epsilon_0} \left(\frac{32\pi^3}{27} \right)^{1/2} \left(\frac{3}{2\pi} \right)^{1/2} \frac{x}{r^3},
 \end{aligned}$$

and after simplification we find

$$\varphi(\mathbf{r}) = \frac{\sigma_0 R^3}{3\epsilon_0} \frac{x}{r^3}, \quad r > R.$$

(c) The interior spherical multipole moments are

$$\begin{aligned} B_{lm} &= \frac{4\pi}{2l+1} \int d^3r' \frac{\rho(\mathbf{r}')}{(r')^{l+1}} Y_{lm}(\Omega') \\ &= \frac{4\pi\sigma_0}{2l+1} \int_0^\infty dr' (r')^{-l+1} \delta(r' - R) \int d\Omega' \sin\theta' \cos\phi' Y_{lm}(\Omega'). \end{aligned}$$

Using Eq. (1), we have that

$$Y_{11}^*(\theta', \phi') - Y_{1,-1}^*(\theta', \phi') = \sqrt{\frac{3}{2\pi}} \sin\theta' \cos\phi',$$

which means

$$B_{lm} = \frac{4\pi\sigma_0}{(2l+1)} R^{-l+1} \sqrt{\frac{2\pi}{3}} (\delta_{1l}\delta_{1m} - \delta_{1l}\delta_{-1,m}).$$

The nonzero moments are B_{11} and $B_{1,-1}$, and are given by

$$B_{11} = \left(\frac{32\pi^3}{27}\right)^{1/2} \sigma_0, \quad B_{1,-1} = -B_{11}.$$

(d) Using the interior multipole expansion, we have

$$\begin{aligned} \varphi(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l B_{lm} r^l Y_{lm}^*(\Omega) \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{32\pi^3}{27}\right)^{1/2} \sigma_0 r [Y_{11}^*(\Omega) - Y_{1,-1}^*(\Omega)] \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{32\pi^3}{27}\right)^{1/2} \sigma_0 r \left(\frac{3}{2\pi}\right)^{1/2} \frac{x}{r}, \end{aligned}$$

and so

$$\varphi(\mathbf{r}) = \frac{\sigma_0 x}{3\epsilon_0}, \quad r < R.$$

(e) The potential expressed in spherical coordinates is

$$\varphi(\mathbf{r}) = \frac{\sigma_0}{3\epsilon_0} \begin{cases} \frac{R^3}{r^2} \sin\theta \cos\phi, & r > R \\ r \sin\theta \cos\phi, & r < R \end{cases}$$

We see that $\varphi(\mathbf{r})$ is continuous at $r = R$, and a direct computation shows that

$$\mathbf{E}(\mathbf{r}) = \frac{\sigma_0}{3\epsilon_0} \begin{cases} \frac{R^3}{r^3} \left(2 \sin \theta \cos \phi \hat{\mathbf{r}} - \cos \theta \cos \phi \hat{\boldsymbol{\theta}} + \sin \phi \hat{\boldsymbol{\phi}} \right), & r > R \\ - \sin \theta \cos \phi \hat{\mathbf{r}} - \cos \theta \cos \phi \hat{\boldsymbol{\theta}} + \sin \phi \hat{\boldsymbol{\phi}}, & r < R \end{cases}$$

Therefore, at $r = R$ we have

$$\hat{\mathbf{r}} \cdot [\mathbf{E}_{\text{out}} - \mathbf{E}_{\text{in}}] = \frac{\sigma_0 \sin \theta \cos \phi}{\epsilon_0},$$

$$\hat{\mathbf{r}} \times [\mathbf{E}_{\text{out}} - \mathbf{E}_{\text{in}}] = 0,$$

which are precisely the matching conditions for the electric field.

(e) Since for $r \gg R$ we have

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{r} + \frac{\mathbf{p} \cdot \mathbf{r}}{r^3} + Q_{ij} \frac{3r_i r_j - r^2 \delta_{ij}}{r^5} + \dots \right],$$

it follows from inspection of part (b) that

$$\mathbf{p} = \frac{4\pi}{3} \sigma_0 R^3 \hat{\mathbf{x}}.$$

Evidently, the “centers” of positive and negative charge are both located along the x axis, and this is to be expected, since the charge distribution is invariant under rotations about the x axis. □

Alternatively, we can recognize that this is just the surface density $\sigma(\theta) = \sigma_0 \cos \theta$ examined in Application 4.3 of Zangwill, but with the axes rotated. To see this, just rotate the distribution counter-clockwise around the y axis to get $x \rightarrow z$ and $z \rightarrow -x$. Then we have $\sigma \rightarrow \sigma_0 z/r = \sigma_0 \cos \theta$, which is an azimuthally symmetric charge density that can be treated in the same way as Problem 4. The multipole moments will be different, but the final potential is the same.

4 (Jackson 4.7.a)

Solution. Since our charge distribution has no azimuthal dependence, at large distances the potential is given by [Eq. (4.95) in Zangwill]

$$\varphi(r, \theta) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{A_l}{r^{l+1}} P_l(\cos \theta). \quad (2)$$

Now, the expansion coefficients A_l are

$$\begin{aligned} A_l &= \int d^3r' (r')^l \rho(r', \theta') P_l(\cos \theta') \\ &= \frac{1}{64\pi} \int d^3r' (r')^l (r')^2 e^{-r'} \sin^2 \theta' P_l(\cos \theta') \\ &= \frac{1}{32} \int_0^\infty dr' (r')^{l+4} e^{-r'} \int_0^\pi d\theta' \sin \theta' \sin^2 \theta' P_l(\cos \theta') \\ &= \frac{1}{32} \int_0^\infty dr' (r')^{l+4} e^{-r'} \int_{-1}^{+1} dx (1 - x^2) P_l(x), \end{aligned}$$

where in the third line we get a 2π from the azimuthal integral, and in the last line we make the substitution $x = \cos \theta'$. Next we note that $P_2(x) = \frac{1}{2}(3x^2 - 1)$ and $P_0(x) = 1$, which means

$$1 - x^2 = \frac{2}{3} [P_0(x) - P_2(x)].$$

Substituting this back into the integral and using the orthogonality relation for Legendre polynomials, we find

$$A_l = \frac{1}{48} \left(2\delta_{l0} - \frac{2}{5}\delta_{l2} \right) \int_0^\infty dr' (r')^{l+4} e^{-r'}.$$

There are only two nonzero expansion coefficients:

$$\begin{aligned} A_0 &= \frac{1}{48} \times 2 \times \int_0^\infty dr' (r')^4 e^{-r'} = 1, \\ A_2 &= \frac{1}{48} \times \left(-\frac{2}{5} \right) \times \int_0^\infty dr' (r')^6 e^{-r'} = -6. \end{aligned}$$

Then since $A_{lm} = A_l \delta_{m0} \sqrt{4\pi/(2l+1)}$, the only nonzero exterior multipole moments are

$$\boxed{A_{00} = \sqrt{4\pi}, \quad A_{20} = -6\sqrt{\frac{4\pi}{5}}.}$$

Meanwhile, substituting the coefficients A_l into Eq. (2), we find

$$\boxed{\begin{aligned} \varphi(r, \theta) &= \frac{1}{4\pi\epsilon_0} \left[\frac{P_0(\cos \theta)}{r} - \frac{6P_2(\cos \theta)}{r^3} \right] \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{9\cos^2 \theta - 3}{r^3} \right]. \end{aligned}}$$

□

1 (Jackson 4.1a–b)

Solution. (a) We will use Zangwill's definition of the exterior spherical multipole moments:

$$A_{lm} = \frac{4\pi}{2l+1} \int d^3r' \rho(\mathbf{r}') (r')^l Y_{lm}^*(\Omega'). \quad (1)$$

From inspection of the figure, we see that

$$\rho(\mathbf{r}) = q \left[\delta(\mathbf{r} - a\hat{\mathbf{x}}) + \delta(\mathbf{r} - a\hat{\mathbf{y}}) - \delta(\mathbf{r} + a\hat{\mathbf{x}}) - \delta(\mathbf{r} + a\hat{\mathbf{y}}) \right].$$

The delta functions make the integration trivial, and we get

$$A_{lm} = \frac{4\pi q a^l}{2l+1} \left[Y_{lm}^*(\hat{\mathbf{x}}) - Y_{lm}^*(-\hat{\mathbf{x}}) + Y_{lm}^*(\hat{\mathbf{y}}) - Y_{lm}^*(-\hat{\mathbf{y}}) \right].$$

Using the parity of the spherical harmonics $Y_{lm}(\hat{\mathbf{r}}) = (-1)^l Y_{lm}(-\hat{\mathbf{r}})$, we conclude

$$A_{lm} = \begin{cases} \frac{8\pi q a^l}{2l+1} [Y_{lm}^*(\hat{\mathbf{x}}) + Y_{lm}^*(\hat{\mathbf{y}})], & l \text{ odd} \\ 0, & l \text{ even} \end{cases}$$

We can see that $A_{lm} = (-1)^l A_{l,-m}^*$, as expected, so we need only provide the multipoles A_{lm} with $m \geq 0$. By direct computation, we find that the first two sets of nonvanishing multipoles are

$$\begin{aligned} A_{11} &= 2qa \sqrt{\frac{2\pi}{3}} (-1 + i), \\ A_{31} &= qa^3 \sqrt{\frac{3\pi}{7}} (1 - i), \\ A_{33} &= -qa^3 \sqrt{\frac{5\pi}{7}} (1 + i). \end{aligned}$$

We check that this is consistent with the Cartesian expansion: an easy calculation shows that the total charge is zero, the dipole moment is $\mathbf{p} = 2qa(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, and the quadrupole tensor is zero. The first and last statements are consistent with the fact that A_{lm} is zero for l even, and some algebra shows that

$$\begin{aligned} \frac{A_{11}Y_{11}(\Omega) + A_{10}Y_{10}(\Omega) + A_{1,-1}Y_{1,-1}(\Omega)}{r^2} &= \frac{2qa \sin \theta (\cos \phi + \sin \phi)}{r^2} \\ &= \frac{2qa(x + y)}{r^3} \\ &= \frac{\mathbf{p} \cdot \mathbf{r}}{r^3}, \end{aligned}$$

as we expect.

(b) A similar computation, but now with

$$\rho(\mathbf{r}) = q \left[\delta(\mathbf{r} - a\hat{\mathbf{z}}) - 2\delta(\mathbf{r}) + \delta(\mathbf{r} + a\hat{\mathbf{z}}) \right].$$

However, if you actually try to compute A_{lm} with this charge density, you'll end up with an A_{00} that is nonzero, which is inconsistent with the fact that the total charge of this system is zero. Instead we can write

$$\rho(\mathbf{r}) = \lim_{\varepsilon \rightarrow 0} q \left[\delta(\mathbf{r} - a\hat{\mathbf{z}}) - \delta(\mathbf{r} - \varepsilon\hat{\mathbf{z}}) + \delta(\mathbf{r} + a\hat{\mathbf{z}}) - \delta(\mathbf{r} + \varepsilon\hat{\mathbf{z}}) \right],$$

and then by Eq. (1) we have

$$A_{lm} = \frac{4\pi q}{2l+1} \lim_{\varepsilon \rightarrow 0} \left[a^l Y_{lm}^*(\hat{\mathbf{z}}) - \varepsilon^l Y_{lm}^*(\hat{\mathbf{z}}) + a^l Y_{lm}^*(-\hat{\mathbf{z}}) - \varepsilon^l Y_{lm}^*(-\hat{\mathbf{z}}) \right].$$

Here we see the significance of the limiting procedure: when $l \neq 0$ we have $\varepsilon^l \rightarrow 0$, but in the case where $l = 0$ we get $\varepsilon^0 \rightarrow 1$, and this causes the expression in brackets to vanish. Thus $A_{00} = 0$, as we expect. Therefore, by the parity of the spherical harmonics we conclude

$$A_{lm} = \begin{cases} \frac{8\pi q a^l}{2l+1} Y_{lm}^*(\hat{\mathbf{z}}), & l \text{ even and } l \neq 0 \\ 0, & l \text{ odd or } l = 0 \end{cases}$$

Again, the property $A_{lm} = (-1)^l A_{l,-m}^*$ is manifest in this expression. Using Mathematica (see next page), the first two sets of nonvanishing multipole moments are

$$\boxed{\begin{aligned} A_{20} &= 4qa^2 \sqrt{\frac{\pi}{5}}, \\ A_{40} &= \frac{4qa^4}{3} \sqrt{\pi}. \end{aligned}}$$

The quadrupole term dominates in the multipole expansion. (This is expected, because the total charge is zero, and the “centers” of positive and negative charge both lie at the origin, so the dipole moment is zero.) $\square$

```

In[ ]:= A[l_?OddQ, m_] :=  $\frac{8 \pi a^l q}{2 l + 1}$  (Conjugate@SphericalHarmonicY[l, m,  $\frac{\pi}{2}$ ,  $\theta$ ] +
      Conjugate@SphericalHarmonicY[l, m,  $\frac{\pi}{2}$ ,  $\frac{\pi}{2}$ ])

In[ ]:= A[1, 0]
      A[1, 1]

Out[ ]:= 0

Out[ ]:=  $(-2 + 2 i) a \sqrt{\frac{2 \pi}{3}} q$ 

In[ ]:= A[3, 0]
      A[3, 1]
      A[3, 2]
      A[3, 3]

Out[ ]:= 0

Out[ ]:=  $(1 - i) a^3 \sqrt{\frac{3 \pi}{7}} q$ 

Out[ ]:= 0

Out[ ]:=  $(-1 - i) a^3 \sqrt{\frac{5 \pi}{7}} q$ 

In[ ]:=  $\frac{A[1, 1] \text{SphericalHarmonicY}[1, 1, \theta, \phi] + A[1, -1] \text{SphericalHarmonicY}[1, -1, \theta, \phi]}{r^2} //$ 
      FullSimplify
Out[ ]:=  $\frac{2 a q \sin[\theta] (\cos[\phi] + \sin[\phi])}{r^2}$ 

```

Figure 1: Mathematica code for Problem 1(a).

```

In[ ]:= A[l_?EvenQ, m_] :=  $\frac{8 \pi q a^l}{2 l + 1}$  Conjugate@SphericalHarmonicY[l, m,  $\theta$ , 0];

In[ ]:= A[2, 2]
      A[2, 1]
      A[2, 0]

Out[ ]:= 0

Out[ ]:= 0

Out[ ]:=  $4 a^2 \sqrt{\frac{\pi}{5}} q$ 

In[ ]:= A[4, 4]
      A[4, 3]
      A[4, 2]
      A[4, 1]
      A[4, 0]

Out[ ]:= 0

Out[ ]:= 0

Out[ ]:= 0

Out[ ]:= 0

Out[ ]:=  $\frac{4}{3} a^4 \sqrt{\pi} q$ 

```

Figure 2: Mathematica code for Problem 1(b).

2 (Zangwill 4.23a)

Solution. The exterior spherical multipole expansion for the potential is

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} \frac{Y_{lm}(\Omega)}{r^{l+1}}, \quad (2)$$

and using this we can write

$$\varphi(R, \Omega) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} \frac{Y_{lm}(\Omega)}{R^{l+1}},$$

where the function on the left is known. Then we multiply by $Y_{l'm'}^*(\Omega)$ and integrate to get

$$\begin{aligned} \int d\Omega' \varphi(R, \Omega') Y_{l'm'}^*(\Omega') &= \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{A_{lm}}{R^{l+1}} \underbrace{\int d\Omega' Y_{lm}(\Omega') Y_{l'm'}^*(\Omega')}_{\delta_{ll'} \delta_{mm'}} \\ &= \frac{1}{4\pi\epsilon_0} \frac{A_{l'm'}}{R^{l'+1}}, \end{aligned}$$

where we've used the orthonormality of the spherical harmonics. Rearranging and dropping the primes on the indices, we get

$$A_{lm} = 4\pi\epsilon_0 R^{l+1} \int d\Omega' \varphi(R, \Omega') Y_{lm}^*(\Omega').$$

Inserting this back into Eq. (2) gives

$$\boxed{\varphi(\mathbf{r}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \left(\frac{R}{r}\right)^{l+1} Y_{lm}(\Omega) \int d\Omega' \varphi(R, \Omega') Y_{lm}^*(\Omega'),}$$

as we wanted to show. For a quick check on this result, we put in the potential due to a point charge, $\varphi(R, \Omega) = q/4\pi\epsilon_0 R$. Then since $Y_{00} = 1/\sqrt{4\pi}$, we have

$$\begin{aligned} \varphi(\mathbf{r}) &= \sum_{l=0}^{\infty} \sum_{m=-l}^l \left(\frac{R}{r}\right)^{l+1} Y_{lm}(\Omega) \frac{q}{4\pi\epsilon_0 R} \underbrace{\sqrt{4\pi} \int d\Omega' Y_{00}(\Omega') Y_{lm}^*(\Omega')}_{\delta_{l0} \delta_{m0}} \\ &= \frac{R}{r} \frac{q}{4\pi\epsilon_0 R} \\ &= \frac{q}{4\pi\epsilon_0 r}, \end{aligned}$$

as we expect. Thus, if you know the potential on a sphere, then you can determine the potential at all points outside of that sphere, even if you are ignorant as to what is actually inside the sphere. (This is a very watered-down version of the statement that the *Dirichlet problem*, where the potential is specified on a closed surface, has a unique solution.) $\square$

1 (Zangwill 5.1)

Solution. Since the cavity is a charge-free region, it follows from Earnshaw's theorem that the electric potential inside the cavity attains its maximum/minimum at the wall of the cavity. But since the cavity is inside a conductor, this wall is an equipotential surface. Thus the maximum and minimum are the same, i.e. the potential is constant everywhere inside the cavity. Then $\mathbf{E} = 0$ inside the cavity, because $\mathbf{E} = -\nabla\varphi$. $\square$

2 (Zangwill 5.5)

Solution. (a) We write the total electrostatic energy due to the total charge distribution $\rho(\mathbf{r}) = \rho_{\text{sp}}(\mathbf{r}) + \rho_{\text{pt}}(\mathbf{r})$ as

$$U_E = U_E[\rho_{\text{sp}}] + \cancel{U_E[\rho_{\text{pt}}]} + \frac{1}{4\pi\epsilon_0} \int d^3r \int d^3r' \frac{\rho_{\text{pt}}(\mathbf{r})\rho_{\text{sp}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad (1)$$

where we've crossed out the term in the middle to indicate that we'll ignore the infinite self-energy of the point charge, and where the third term is the interaction energy V_E . First we compute the self-energy of the sphere,

$$U_E[\rho_{\text{sp}}] = \frac{1}{2} \int d^3r \rho_{\text{sp}}(\mathbf{r}) \varphi_{\text{sp}}(\mathbf{r}). \quad (2)$$

The charge density is given by

$$\rho_{\text{sp}}(r, \theta) = \delta(r - R) \sum_{l=1}^{\infty} \sigma_l P_l(\cos \theta),$$

and for the potential we use the (azimuthally symmetric) exterior multipole expansion,

$$\varphi_{\text{sp}}(r, \theta) = \frac{1}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{A_l}{r^{l+1}} P_l(\cos \theta).$$

The expansion coefficients are computed as follows:

$$\begin{aligned} A_l &= \int d^3r' r'^l \rho(r', \theta') P_l(\cos \theta') \\ &= 2\pi R^{l+2} \int_{-1}^{+1} d(\cos \theta') \left(\sum_{l'=1}^{\infty} \sigma_{l'} P_{l'}(\cos \theta') \right) P_l(\cos \theta') \\ &= 2\pi R^{l+2} \sum_{l'=1}^{\infty} \sigma_{l'} \int_{-1}^{+1} d(\cos \theta') P_{l'}(\cos \theta') P_l(\cos \theta') \\ &= 2\pi R^{l+2} \sum_{l'=1}^{\infty} \sigma_{l'} \frac{2}{2l+1} \delta_{ll'}. \end{aligned}$$

This shows that

$$A_l = \begin{cases} \frac{4\pi}{2l+1} \sigma_l R^{l+2}, & l \neq 0 \\ 0, & l = 0 \end{cases}$$

Substituting this back into the multipole expansion, we obtain

$$\varphi_{\text{sp}}(r, \theta) = \frac{1}{\epsilon_0} \sum_{l=1}^{\infty} \frac{\sigma_l R}{2l+1} \left(\frac{R}{r} \right)^{l+1} P_l(\cos \theta').$$

Notice that we've changed the lower limit on the sum to $l = 1$, since $A_0 = 0$. We put this back into Eq. (2) to compute the sphere self-energy:

$$\begin{aligned}
 U_E[\rho_{\text{sp}}] &= \frac{1}{2} \int d^3r \left[\delta(r - R) \sum_{l'=1}^{\infty} \sigma_{l'} P_{l'}(\cos \theta) \right] \left[\frac{1}{\epsilon_0} \sum_{l=1}^{\infty} \frac{\sigma_l R}{2l+1} \left(\frac{R}{r} \right)^{l+1} P_l(\cos \theta') \right] \\
 &= \frac{1}{2\epsilon_0} \sum_{l'=1}^{\infty} \sum_{l=1}^{\infty} \frac{\sigma_{l'} \sigma_l R}{2l+1} \int d^3r \delta(r - R) \left(\frac{R}{r} \right)^{l+1} P_{l'}(\cos \theta) P_l(\cos \theta) \\
 &= \frac{1}{2\epsilon_0} \sum_{l'=1}^{\infty} \sum_{l=1}^{\infty} \frac{\sigma_{l'} \sigma_l R}{2l+1} 2\pi R^2 \int_{-1}^{+1} d(\cos \theta) P_{l'}(\cos \theta) P_l(\cos \theta) \\
 &= \frac{1}{2\epsilon_0} \sum_{l'=1}^{\infty} \sum_{l=1}^{\infty} \frac{\sigma_{l'} \sigma_l R}{2l+1} 2\pi R^2 \frac{2}{2l+1} \delta_{ll'} \\
 &= \frac{1}{\epsilon_0} \sum_{l=1}^{\infty} \frac{\sigma_l}{2l+1} \left[\frac{\sigma_l R^3}{2} \frac{4\pi}{2l+1} \right].
 \end{aligned}$$

Meanwhile, the interaction energy is¹

$$\begin{aligned}
 V_E &= \frac{1}{4\pi\epsilon_0} \int d^3r \int d^3r' \frac{q \delta(\mathbf{r} - r\hat{\mathbf{z}}) \delta(r' - R)}{|\mathbf{r} - \mathbf{r}'|} \sum_{l=1}^{\infty} \sigma_l P_l(\cos \theta') \\
 &= \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{q \delta(r' - R)}{|r\hat{\mathbf{z}} - \mathbf{r}'|} \sum_{l=1}^{\infty} \sigma_l P_l(\cos \theta') \\
 &= \sum_{l=1}^{\infty} \frac{q\sigma_l R^2}{2\epsilon_0} \int_{-1}^{+1} d(\cos \theta') \frac{1}{|r\hat{\mathbf{z}} - R\hat{\mathbf{r}}'|} P_l(\cos \theta').
 \end{aligned}$$

Recalling that $r > R$, we use (4.78) in Zangwill to write

$$\frac{1}{|r\hat{\mathbf{z}} - R\hat{\mathbf{r}}'|} = \frac{1}{r} \sum_{l=0}^{\infty} \left(\frac{R}{r} \right)^l P_l(\underbrace{\hat{\mathbf{r}}' \cdot \hat{\mathbf{z}}}_{\cos \theta'}),$$

so that

$$\begin{aligned}
 V_E &= \sum_{l=1}^{\infty} \sum_{l'=0}^{\infty} \frac{q\sigma_l R^2}{2\epsilon_0} \frac{1}{r} \left(\frac{R}{r} \right)^{l'} \int_{-1}^{+1} d(\cos \theta') P_{l'}(\cos \theta') P_l(\cos \theta') \\
 &= \sum_{l=1}^{\infty} \sum_{l'=0}^{\infty} \frac{q\sigma_l R^2}{2\epsilon_0} \frac{1}{r} \left(\frac{R}{r} \right)^{l'} \frac{2}{2l+1} \delta_{ll'} \\
 &= \sum_{l=1}^{\infty} \frac{q\sigma_l R^2}{2\epsilon_0} \frac{1}{r} \left(\frac{R}{r} \right)^l \frac{2}{2l+1},
 \end{aligned}$$

¹In the first line, we write $\delta(\mathbf{r} - r\hat{\mathbf{z}})$, which is confusing, because the r in $r\hat{\mathbf{z}}$ denotes the distance from the point charge to the center of the sphere, and *not* the magnitude of the vector $\mathbf{r}$. Fortunately, this delta function gets integrated out in the next line, so there is no further chance of confusion.

and a bit of simplifying gives

$$V_E = \frac{1}{\epsilon_0} \sum_{l=1}^{\infty} \frac{\sigma_l}{2l+1} \frac{qR^{l+2}}{r^{l+1}}.$$

Finally, putting everything back into Eq. (1), we get a total electrostatic energy of

$$U_E = \frac{1}{\epsilon_0} \sum_{l=1}^{\infty} \frac{\sigma_l}{2l+1} \left[\frac{\sigma_l R^3}{2} \frac{4\pi}{2l+1} + \frac{qR^{l+2}}{r^{l+1}} \right],$$

as was to be shown. Some printings of Zangwill have a lower limit of $l = 0$ in the sum, which is an error; see the errata available online. [This is straightforward to see from the fact that our definition of $\sigma(\theta)$ does not include a σ_0 term.]

(b) Thomson's theorem says that for a body of fixed shape, size, and charge, the charge density $\rho(\mathbf{r})$ is distributed in a way that *minimizes* the total electrostatic energy U_E . Thus, the coefficients σ_l in the expansion

$$\rho_{\text{sp}}(r, \theta) = \delta(r - R) \sum_{l=1}^{\infty} \sigma_l P_l(\cos \theta)$$

are such that U_E is a minimum. Taking the gradient of U_E with respect to these coefficients and equating the resulting vector to zero, we find for each $l \geq 1$

$$\begin{aligned} 0 &= \frac{1}{2l+1} \left[\frac{\sigma_l R^3}{2} \frac{4\pi}{2l+1} + \frac{qR^{l+2}}{r^{l+1}} \right] + \frac{\sigma_l}{2l+1} \frac{R^3}{2} \frac{4\pi}{2l+1} \\ &= \frac{2\pi\sigma_l R^3}{(2l+1)^2} + \frac{qR^{l+2}}{(2l+1)r^{l+1}} + \frac{2\pi\sigma_l R^3}{(2l+1)^2} \\ &= \frac{4\pi R^3}{(2l+1)^2} \sigma_l + \frac{qR^{l+2}}{(2l+1)r^{l+1}}. \end{aligned}$$

Rearranging and simplifying, we get

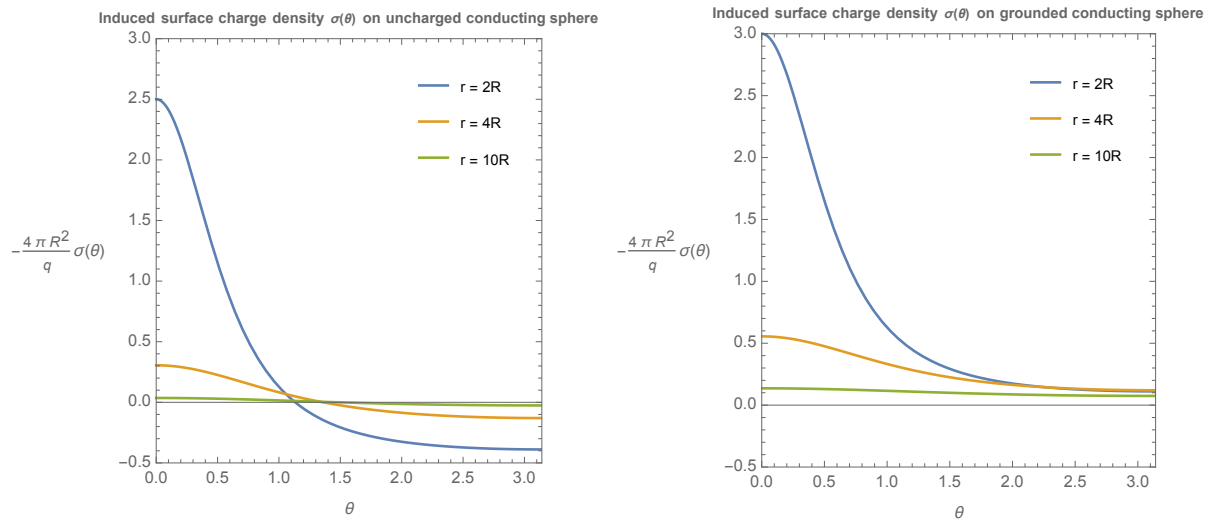
$$\sigma_l = -(2l+1) \frac{q}{4\pi R^2} \left(\frac{R}{r} \right)^{l+1},$$

and by substituting back into our expression for $\sigma(\theta)$, we obtain

$$\sigma(\theta) = \sum_{l=1}^{\infty} -(2l+1) \frac{q}{4\pi R^2} \left(\frac{R}{r} \right)^{l+1} P_l(\cos \theta).$$

Note that due to the orthogonality of the Legendre polynomials, the total induced charge $Q = \int \sigma dA$ is zero, as we expect. To do the same problem with a *charged* sphere, we retain the $l = 0$ term in our expansion to write $\sigma(\theta) = \sum_{l=0}^{\infty} \sigma_l P_l(\cos \theta)$, and then we find that σ_0 is fixed by the total charge on the sphere. (If we do this, then all of the sums that we write will inherit the lower bound of $l = 0$.)

It's interesting to compare this answer against the induced surface charge density on a *grounded* conducting sphere, which is not constrained to have zero charge. The plot on the left is our result, using the first 100 terms in the expansion, while the plot on the right uses (2.5) in Jackson:



In both plots, the induced surface charge density is negative on the side closest to the point charge, and more negative charge accumulates there as the point charge gets closer to the sphere. Notice that the grounded conducting sphere always picks up a net negative charge, whereas the uncharged conducting sphere has a surface charge density that is positive on the side of the sphere opposite the point charge. $\square$

3 (Zangwill 5.9)

Solution. (a)² We know, from Application 4.3 in Zangwill, that the surface charge density $\sigma(\theta) = \sigma_0 \cos \theta$ on a sphere of radius R produces an electric field that is exactly dipolar for $r > R$. The idea is to place this charge density on the wall of the spherical cavity, and then choose σ_0 so that the dipole field from the sphere cancels out exactly with the field from the point dipole. Then we have $\mathbf{E} = 0$ in the conductor, as required.

Now we proceed as described. The electric field due to a point dipole at the origin, pointing in the $\hat{\mathbf{z}}$ direction, is

$$\mathbf{E}_D(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}).$$

Next, by using the potential arising from the surface charge density $\sigma(\theta) = \sigma_0 \cos \theta$ distributed over a sphere of radius R (see Application 4.3 in Zangwill), we get

$$\mathbf{E}_{\text{sp}}(r, \theta) = \begin{cases} \frac{\sigma_0}{3\epsilon_0} (-\cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}), & r < R \\ \frac{\sigma_0 R^3}{3\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}), & r > R \end{cases} \quad (3)$$

Thus, if we want the total electric field $\mathbf{E} = \mathbf{E}_D + \mathbf{E}_{\text{sp}}$ to be zero for $r > R$, then we need

$$\frac{p}{4\pi\epsilon_0 r^3} + \frac{\sigma_0 R^3}{3\epsilon_0 r^3} = 0,$$

which is satisfied when

$$\sigma_0 = -\frac{3p}{4\pi R^3}.$$

We conclude that the induced surface charge density on the wall of the cavity is

$$\sigma(\theta) = -\frac{3p}{4\pi R^3} \cos \theta.$$

(b) This follows directly from Eq. (4.21) in Zangwill: the force exerted on the point dipole by the charge induced on the sphere is

$$\mathbf{F} = \nabla(\mathbf{p} \cdot \mathbf{E}_{\text{sp}})|_{\mathbf{r}=0} = 0,$$

because $\mathbf{E}_{\text{sp}}$ is constant everywhere inside the sphere. (This isn't obvious from the way we've written Eq. (3), but it's straightforward to show that $-\cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}} = -\hat{\mathbf{z}}$.) Thus $\mathbf{p} \cdot \mathbf{E}_{\text{sp}}$ is a constant inside the sphere, so the gradient of this dot product vanishes at the location of the point dipole. $\square$

²I'd like to thank Lorenzo Joquino for the suggestion to look at Zangwill 4.20.

1 (Zangwill 7.1)

Solution. For the region enclosed by S , the general solution to Laplace's equation is

$$\varphi(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} r^l Y_{lm}(\theta, \phi). \quad (1)$$

We discard the part that goes as $r^{-(l+1)}$ because it blows up at the origin.

(a) Integrating over the surface S , we have

$$\begin{aligned} \int_S dS \varphi(\mathbf{r}) &= \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} R^{l+2} \int d\Omega Y_{lm}(\Omega) \\ &= \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} R^{l+2} \sqrt{4\pi} \int d\Omega Y_{lm}(\Omega) Y_{00}^*(\Omega) \\ &= \sqrt{4\pi} A_{00} R^2, \end{aligned}$$

where we've used $Y_{00}(\Omega) = 1/\sqrt{4\pi}$ and the orthogonality of the spherical harmonics. Referring back to Eq. (1), we see that the value of the potential at the origin is just

$$\varphi(0) = A_{00} Y_{00} = \frac{1}{\sqrt{4\pi}} A_{00},$$

and substituting this into our result for the surface integral yields

$$\boxed{\int_S dS \varphi(\mathbf{r}) = 4\pi R^2 \varphi(0).}$$

(b) Since $z = r \cos \theta$, we have

$$\begin{aligned} \int_S dS z \varphi(\mathbf{r}) &= \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{lm} R^{l+3} \sqrt{\frac{4\pi}{3}} \int d\Omega Y_{lm}(\Omega) Y_{10}^*(\Omega) \\ &= \sqrt{\frac{4\pi}{3}} A_{10} R^4, \end{aligned}$$

where we've used the fact that $Y_{10}(\Omega) = \sqrt{3/4\pi} \cos \theta$. Next, we write out the terms in (1) up to $l = 1$, and then transform to Cartesian coordinates with the aid of Mathematica:

$$\begin{aligned} \varphi(\mathbf{r}) &= A_{00} Y_{00} + (A_{1,-1} Y_{1,-1} + A_{10} Y_{10} + A_{11} Y_{11}) r + \dots \\ &= \frac{A_{00}}{\sqrt{4\pi}} + \left(\sqrt{\frac{3}{4\pi}} z A_{10} + \sqrt{\frac{3}{8\pi}} [(x - iy) A_{1,-1} - (x + iy) A_{11}] \right) + \dots \end{aligned}$$

Taking the partial derivative with respect to z and then evaluating at $\mathbf{r} = 0$, the first term vanishes, and all terms beyond what we've written vanish when evaluated at the origin, since the higher-order spherical harmonics go like $(x - iy)^2$, $(x - iy)z$, and so on.¹ Thus

$$\left. \frac{\partial \varphi}{\partial z} \right|_{\mathbf{r}=0} = \sqrt{\frac{3}{4\pi}} A_{10},$$

and by comparing against the surface integral we computed, we see that

$$\int_S dS z \varphi(\mathbf{r}) = \frac{4\pi}{3} R^4 \left. \frac{\partial \varphi}{\partial z} \right|_{\mathbf{r}=0}$$

as we wanted to show. □

```
In[12]:= TransformedField["Spherical" -> "Cartesian",
(
  A[1, -1] SphericalHarmonicY[1, -1,  $\theta$ ,  $\phi$ ]
  + A[1, 0] SphericalHarmonicY[1, 0,  $\theta$ ,  $\phi$ ]
  + A[1, 1] SphericalHarmonicY[1, 1,  $\theta$ ,  $\phi$ ]
) r,
{r,  $\theta$ ,  $\phi$ } -> {x, y, z}
] // FullSimplify

Out[12]=  $\frac{1}{4} \sqrt{\frac{3}{\pi}} (2 z A[1, 0] + \sqrt{2} ((x - i y) A[1, -1] - (x + i y) A[1, 1]))$ 
```

¹This isn't a rigorous argument; I convinced myself of this by squinting at a table of spherical harmonics expressed in Cartesian coordinates.

2 (Jackson 3.1)

Solution. We start from the general solution to Laplace's equation for azimuthal symmetry, written in the form

$$\varphi(r, \theta) = \sum_{\ell=0}^{\infty} \left[A_{\ell} \left(\frac{a}{r} \right)^{\ell+1} + B_{\ell} \left(\frac{r}{b} \right)^{\ell} \right] P_{\ell}(\cos \theta). \quad (2)$$

From the boundary condition on the inner shell, we obtain

$$\sum_{\ell} \left[A_{\ell} + B_{\ell} \left(\frac{a}{b} \right)^{\ell} \right] \int_0^{\pi} d\theta \sin \theta P_{\ell}(\cos \theta) P_{\ell'}(\cos \theta) = \int_0^{\pi} d\theta \sin \theta V_{\text{in}}(\theta) P_{\ell'}(\cos \theta).$$

Using the orthogonality property on the left-hand side (there is an overall minus sign, due to the bounds), we get

$$\begin{aligned} A_{\ell} + B_{\ell} \left(\frac{a}{b} \right)^{\ell} &= -\frac{2\ell+1}{2} V \int_0^{\pi/2} d\theta \sin \theta P_{\ell}(\cos \theta) \\ &= \frac{2\ell+1}{2} V \int_0^1 dx P_{\ell}(x). \end{aligned}$$

Similarly, we find that the boundary condition on the outer shell implies

$$\begin{aligned} A_{\ell} \left(\frac{a}{b} \right)^{\ell+1} + B_{\ell} &= -\frac{2\ell+1}{2} V \int_{\pi/2}^{\pi} d\theta \sin \theta P_{\ell}(\cos \theta) \\ &= -\frac{2\ell+1}{2} V \int_0^{-1} dx P_{\ell}(x) \\ &= (-1)^{\ell} \frac{2\ell+1}{2} V \int_0^1 dx P_{\ell}(x), \end{aligned}$$

where in the second-to-last line we use the property $P_{\ell}(-x) = (-1)^{\ell} P_{\ell}(x)$. Recognizing that the same integral appears in both boundary conditions, we define

$$I_{\ell} \equiv \frac{2\ell+1}{2} \int_0^1 dx P_{\ell}(x), \quad (3)$$

so that we have the pair of equations

$$\begin{aligned} A_{\ell} + B_{\ell} \left(\frac{a}{b} \right)^{\ell} &= V I_{\ell}, \\ A_{\ell} \left(\frac{a}{b} \right)^{\ell+1} + B_{\ell} &= (-1)^{\ell} V I_{\ell}, \end{aligned}$$

which has solution

$$A_{\ell} = \frac{1 + (-1)^{\ell+1} (a/b)^{\ell}}{1 - (a/b)^{2\ell+1}} V I_{\ell}, \quad (4)$$

$$B_{\ell} = \frac{(a/b)^{\ell+1} + (-1)^{\ell+1}}{(a/b)^{2\ell+1} - 1} V I_{\ell}. \quad (5)$$

Substituting these coefficients back into Eq. (2) and noting that from the definitions (3)–(5) we have $A_0 = 0$ and $B_0 = VI_0 = V/2$, we get

$$\varphi(r, \theta) = V \left\{ \frac{1}{2} + \sum_{\ell=1}^{\infty} \left[\frac{1 + (-1)^{\ell+1} \epsilon^{\ell}}{1 - \epsilon^{2\ell+1}} \left(\frac{a}{r} \right)^{\ell+1} + \frac{\epsilon^{\ell+1} + (-1)^{\ell+1}}{\epsilon^{2\ell+1} - 1} \left(\frac{r}{b} \right)^{\ell} \right] I_{\ell} P_{\ell}(\cos \theta) \right\},$$

where we've defined $\epsilon \equiv a/b$ for simplicity. It only remains for us to evaluate the integral I_{ℓ} . We quote the result from Gradshteyn and Ryzhik 7th ed., 7.113–1:

$$I_{\ell} = (2\ell + 1) \frac{\sin\left(\frac{\pi\ell}{2}\right)}{\pi\Lambda\ell(\ell + 1)} \quad \text{for } \ell \geq 1, \text{ where } \Lambda \equiv \frac{\Gamma(\frac{\ell+1}{2})}{\Gamma(\frac{1}{2})\Gamma(1 + \frac{\ell}{2})}.$$

For reference, we collect the values of I_{ℓ} up to $\ell = 15$ in a table:

```
In[23]:= gradI[l_] := If[l == 0, 1/2, (2 l + 1) Sin[Pi l/2] Gamma[1/2] Gamma[1 + l/2] / (pi l (l + 1) Gamma[l/2 + 1])

In[24]:= Grid[Transpose@Table[{l, gradI[l]}, {l, 0, 15}], Frame -> All]

Out[24]=
```

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1/2	3/4	0	-7/16	0	11/32	0	-75/256	0	133/512	0	-483/2048	0	891/4096	0	-13299/65536

Noting that $I_{\ell} = 0$ for ℓ even and $\ell \neq 0$, we may simplify our expression further:

$$\varphi(r, \theta) = V \left\{ \frac{1}{2} + \sum_{\ell=1}^{\infty} \left[\frac{1 + \epsilon^{\ell}}{1 - \epsilon^{2\ell+1}} \left(\frac{a}{r} \right)^{\ell+1} + \frac{1 + \epsilon^{\ell+1}}{\epsilon^{2\ell+1} - 1} \left(\frac{r}{b} \right)^{\ell} \right] I_{\ell} P_{\ell}(\cos \theta) \right\} \quad (\epsilon \equiv a/b)$$

This potential satisfies the boundary conditions and Laplace's equation in the region $a \leq r \leq b$, so we've found our solution. As a check, we consider two limiting cases. In the limit $a \rightarrow 0$, we have $\epsilon \rightarrow 0$, we may assume a/r is negligible, and $B_{\ell} \rightarrow -VI_{\ell}$. Using our table, we find

$$\lim_{a \rightarrow 0} \varphi(r, \theta) = V \left[\frac{1}{2} - \frac{3r}{4b} P_1(\cos \theta) + \frac{7}{16} \left(\frac{r}{b} \right)^3 P_3(\cos \theta) - \frac{11}{32} \left(\frac{r}{b} \right)^5 P_5(\cos \theta) + \dots \right].$$

Similarly, in the $b \rightarrow \infty$ limit, we have $\epsilon \rightarrow 0$, r/b is negligible, and $A_{\ell} \rightarrow VI_{\ell}$, so we get

$$\lim_{b \rightarrow \infty} \varphi(r, \theta) = V \left[\frac{1}{2} + \frac{3}{4} \left(\frac{a}{r} \right)^2 P_1(\cos \theta) - \frac{7}{16} \left(\frac{a}{r} \right)^4 P_3(\cos \theta) + \frac{11}{32} \left(\frac{a}{r} \right)^6 P_5(\cos \theta) + \dots \right].$$

Both of these expressions are consistent with the solution (3.36) in Jackson, where we put $V \rightarrow \pm V/2$ and add on a constant $V/2$ everywhere to satisfy the boundary condition. (In both of these limits, there is effectively one boundary condition; one shell is either negligibly small or is located out at infinity.)

As a final, visual check, we plot this solution along the x - z plane. We see that the boundary conditions are close to being satisfied on both shells, with an overshoot at the discontinuity which is expected of a Fourier series expansion.

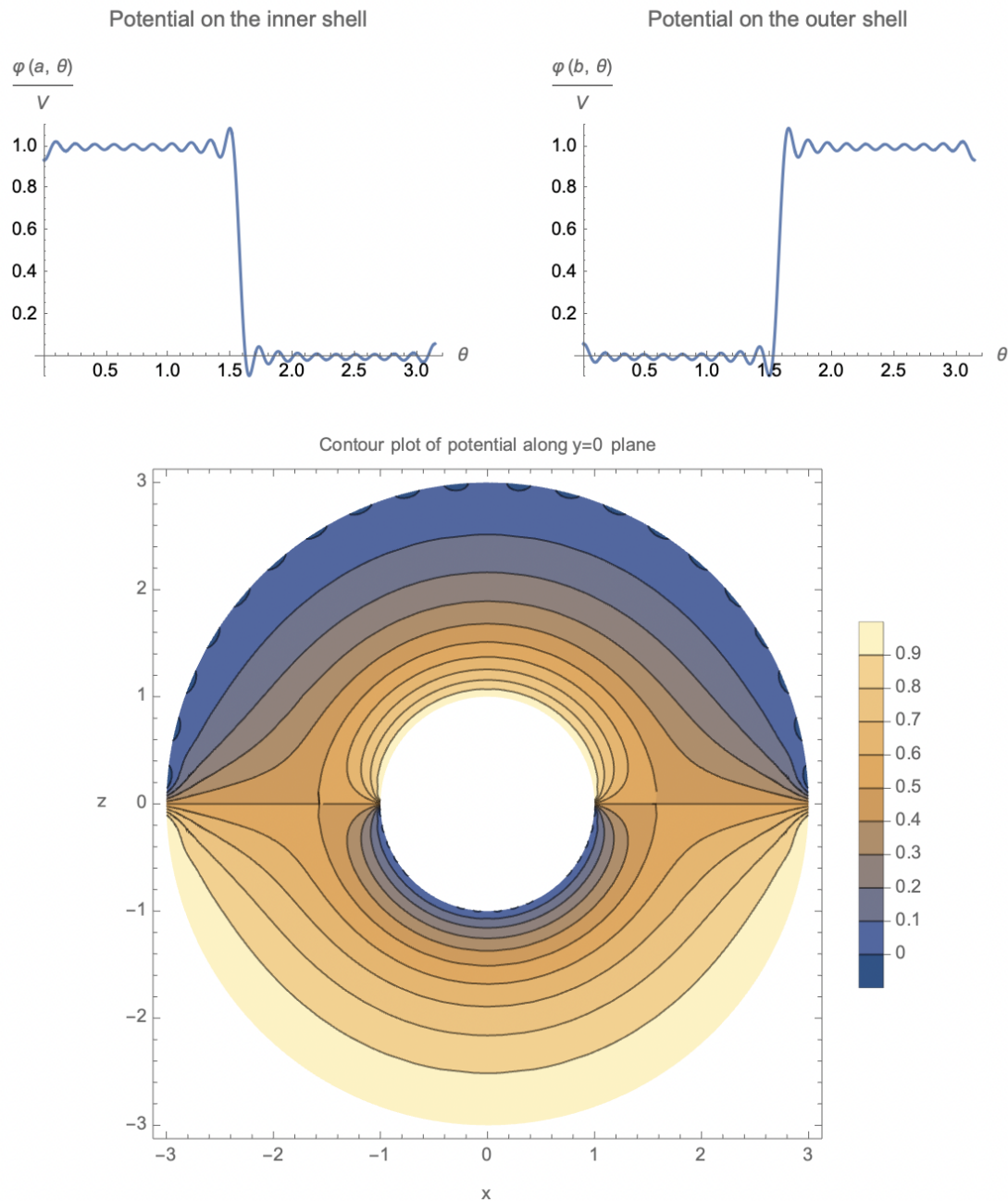


Figure 1: A solution with $a = 1$ and $b = 3$ using terms up to $\ell = 40$ in the expansion.

□

Jackson 3.1

```

In[ ]:= A[l_] := 
$$\frac{\text{Gamma}\left[\frac{l+1}{2}\right]}{\text{Gamma}\left[1/2\right] \text{Gamma}\left[1+\frac{l}{2}\right]}$$
;

gradI[l_] := If[l == 0,  $\frac{1}{2}$ , (2 l + 1)  $\frac{\text{Sin}\left[\frac{\pi l}{2}\right]}{\pi A[l] l (l+1)}$ ];

analyticA[l_] := 
$$\frac{1 + (-1)^{l+1} (a/b)^l}{1 - (a/b)^{1+2l}}$$
 gradI[l];

analyticB[l_] := 
$$\frac{(a/b)^{1+l} + (-1)^{l+1}}{(a/b)^{1+2l} - 1}$$
 gradI[l];

(* compute coefficients up to l=40 *)
lmax = 40;

analyticAs = Table[analyticA[l], {l, 0, lmax}];
analyticBs = Table[analyticB[l], {l, 0, lmax}];

In[ ]:= 
$$\phi[r_, \theta_] := \text{Sum}\left[\left(\text{analyticAs}[[l+1]] \left(\frac{a}{r}\right)^{l+1} + \text{analyticBs}[[l+1]] \left(\frac{r}{b}\right)^l\right) \text{LegendreP}[l, \text{Cos}[\theta]], \{l, 0, \text{Length}[\text{analyticAs}] - 1\}\right];$$

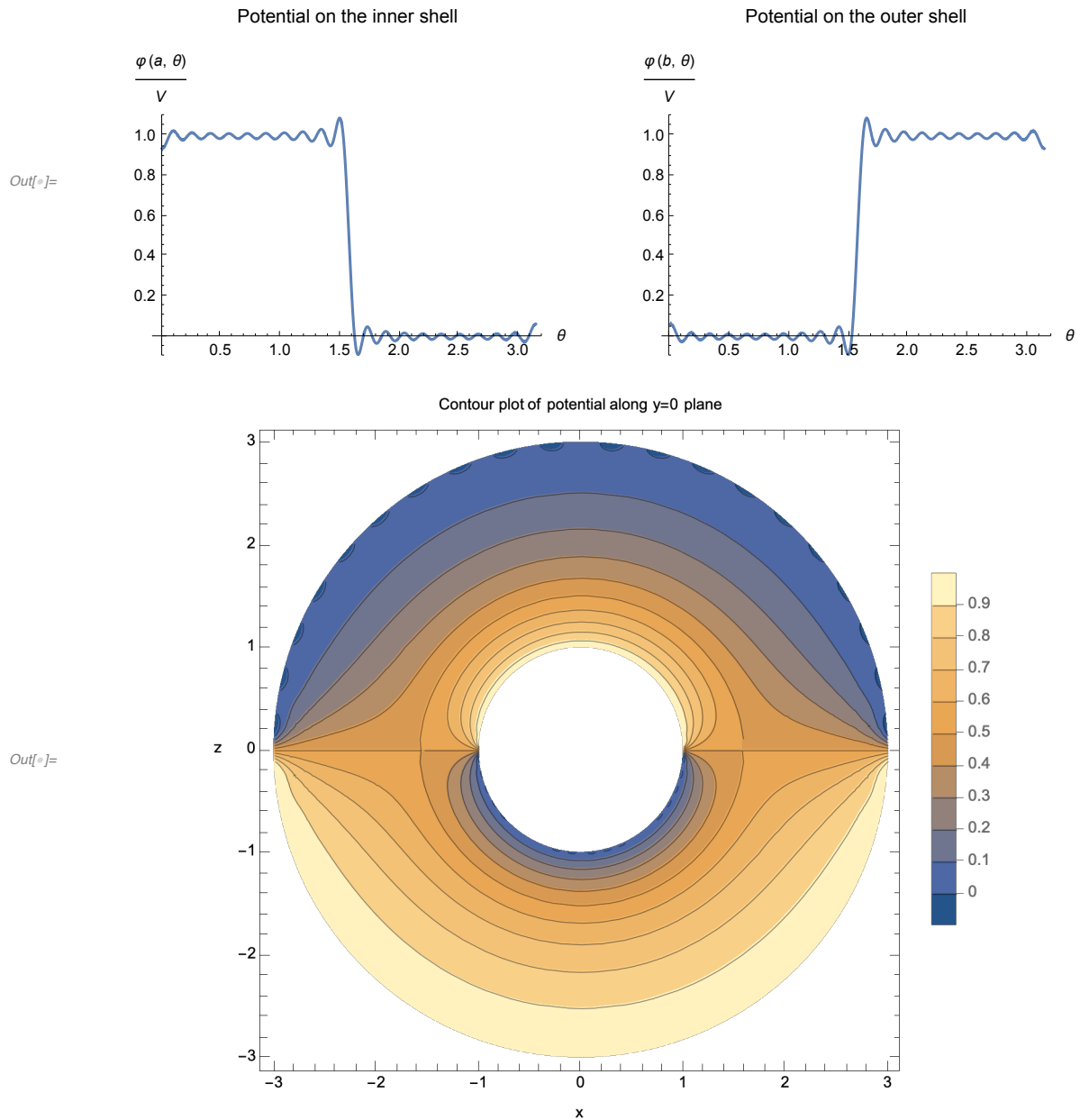

In[ ]:= a = 1;
b = 3;
GraphicsRow[{{Plot[ $\phi[a, \theta]$  // Evaluate, { $\theta$ , 0,  $\pi$ }, PlotRange → {-0.1, 1.1},
  PlotLabel → "Potential on the inner shell", AxesLabel → {" $\theta$ ", " $\frac{\phi(a, \theta)}{V}$ "}}],
  Plot[ $\phi[b, \theta]$  // Evaluate, { $\theta$ , 0,  $\pi$ }, PlotRange → {-0.1, 1.1},
  PlotLabel → "Potential on the outer shell", AxesLabel → {" $\theta$ ", " $\frac{\phi(b, \theta)}{V}$ "}}}
]

cp1 = ContourPlot[ $\phi[r, t]$  // Evaluate,
  {r, a, b}, {t, - $\pi$ ,  $\pi$ }, Contours → 10, PlotRange → All,
  RegionFunction → Function[{r},  $a^2 < r^2 < b^2$ ], PlotLegends → Automatic];

cp2 = Show[cp1 /. GraphicsComplex[c_, rest___] :>
  GraphicsComplex[c /. {a_, b_} :> (a {Sin[b], Cos[b]}), rest],
  AspectRatio → 1, ImageSize → 400, FrameLabel →
  {"x", Rotate["z",  $\pi/2$ ], "Contour plot of potential along y=0 plane"}];

cp2

```



Tabulating l_i up to $l = 15$

$$\text{In[]:= gradI[l_] := If[l == 0, \frac{1}{2}, (2l+1) \frac{\sin\left[\frac{\pi l}{2}\right]}{\pi l(l+1)} \frac{\Gamma\left[\frac{1}{2}\right] \Gamma\left[1 + \frac{l}{2}\right]}{\Gamma\left[\frac{l+1}{2}\right]}]$$

Out[]:= Grid[Transpose@Table[{l, gradI[l]}, {l, 0, 15}], Frame → All]

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\frac{1}{2}$	$\frac{3}{4}$	0	$-\frac{7}{16}$	0	$\frac{11}{32}$	0	$-\frac{75}{256}$	0	$\frac{133}{512}$	0	$-\frac{483}{2048}$	0	$\frac{891}{4096}$	0	$-\frac{13299}{65536}$

Three equivalent expressions for I_l

$$ln[\bullet] := (* \text{ Jackson (3.26) } *)$$

$$\text{jackI}[\text{l}_-] := \text{If}[\text{Mod}[\text{l}, 2] == 1, \frac{1}{2} \left(-\frac{1}{2}\right)^{(\text{l}-1)/2} \frac{(2\text{l}+1)(\text{l}-2)!!}{2\left(\frac{\text{l}+1}{2}\right)!}, 0]$$

(* Gradshteyn and Ryzhik 7e 7.113-1 *)

$$\text{gradI}[\text{l}_-] := (2 \text{l} + 1) \frac{\text{Sin}\left[\frac{\pi \text{l}}{2}\right]}{\pi \text{l} (\text{l} + 1)} \frac{\text{Gamma}\left[\frac{1}{2}\right] \text{Gamma}\left[1 + \frac{\text{l}}{2}\right]}{\text{Gamma}\left[\frac{\text{l} + 1}{2}\right]}$$

(* via the Rodrigues formula *)

$$\text{rodrI}[\ell_]:= \text{If}[\text{Mod}[\ell, 2] == 1, \frac{2\ell+1}{2} (-1)^{(\ell-1)/2} \frac{(\ell-1)!}{2^\ell \ell!} \text{Binomial}[\ell, \frac{\ell-1}{2}], 0]$$

```
In[•]:= Table[jackI[l] - gradI[l], {l, 1, 100}]
```

```
Table[gradI[l] - rodrI[l], {l, 1, 100}]
```

```
Table[rodrI[l] - jackI[l], {l, 1, 100}]
```

[illegible][illegible][illegible]